

ERP Data Quality & Business Intelligence Analysis - Mini Portfolio

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Business Problem

Poor ERP master data quality was affecting reporting reliability, operational efficiency, and decision-making.

Data Analysed

- 7,454 ERP data quality issues
- 17,000+ master data records
- Product master data
- Customer master data
- ERP error reports
- User survey data (26 respondents)

Tools

- Power BI
- SQL concepts
- ERP Data Quality Reports
- Excel

Key Findings

- ✓ 82% of identified issues related to data completeness
- ✓ 6,355 high-severity issues identified
- ✓ Data quality problems concentrated in a single operational site
- ✓ Missing customer and origin data negatively impacted reporting and traceability

DQ REPORT ANALYSIS

Power BI Project | Dinesh Welikala

Key Insight: Completeness errors dominate data quality issues | Most problems are concentrated in a small number of sites.

7454

Total DQ Issues Identified

6635

High Severity DQ issues (Critical Risk)

819

Medium Severity DQ issues

1.69

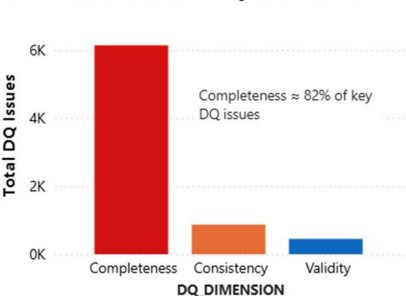
Avg DQ issue per product

0.59

Site error concentration index

> 0.50 = High risk
0.30-0.50 = Medium risk
< 0.30 = Low risk

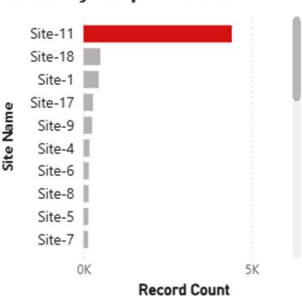
Root Cause: DQ Issues by Dimension



Business Impact

- Increased manual correction effort
- Delayed reporting cycles
- Reduced decision reliability
- Higher operational inefficiency

Errors by Responsible Site



Immediate Recommended Action

- Focus data correction on Site-11
- Improve completeness validation rules
- Strengthen master data entry controls

Top Data Quality Failures Driving Errors

FILE_NAME	DQ_DIMENSION	SEVERITY	Total DQ Issues	% Error Contribution
Missing Country of Origin	Completeness	High	2835	38.03 %
Missing Cust Stat Number	Completeness	High	2496	33.49 %
Wrong Package Type	Consistency	Medium	524	7.03 %
Missing GTIN	Completeness	High	506	6.79 %
Asset class and Part Status Mismatch	Consistency	High	345	4.63 %
Total			6706	89.97 %

Priority Actions

- Site-11 (primary focus point)
- Implement mandatory field validation
- Strengthen master data governance

Business Impact Identified

- Increased manual correction effort
- Reduced reporting reliability
- Delayed reporting cycles
- Lower confidence in business decisions
- Higher operational inefficiency

Recommendations

- Strengthen master data governance
- Implement mandatory field validation
- Define ownership responsibilities
- Introduce continuous data quality monitoring
- Improve consistency of master data entry

Skills Demonstrated

Business

- ERP Process Understanding
- Business Process Analysis
- Operational Reporting

Analytics

- Power BI
- SQL Analysis
- Root Cause Analysis
- Data Quality Assessment

Governance

- Master Data Governance
- Data Validation
- Data Quality Monitoring

Outcome

Identified critical ERP data quality risks affecting reporting reliability and business operations, and proposed governance-focused solutions to improve decision-making and operational performance. Proposed practical corrective actions.